

Assignment No	11
Title	Number Guessing App
Objective	Creating a number guessing app using scripting elements and implicit object methods.
Roll No	MCA2511

Code:

Index.jsp

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1"%>
<!DOCTYPE html>
<html>
<body>
<form action="NextGuess.jsp">
    <h1>Number Guesser</h1>
    <h2>Select a Number between 0 and 100</h2>
    <input type="submit" name="btnSub" value="Start"/>
    <input type="hidden" name="low" value="0"/>
    <input type="hidden" name="high" value="100"/>
    <input type="hidden" name="cnt" value="0">
</form>
</body>
</html>
```

NextGuess.jsp

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1"%>
<!DOCTYPE html>
<html>
<body>
    <%-- <1> Declare Guess variable --%>
    <%! int guess; %>
```

```
<%-- <2> Fetch all variables --%>
<%
    int low = Integer.parseInt(request.getParameter("low"));
    int high = Integer.parseInt(request.getParameter("high"));
    int cnt = Integer.parseInt(request.getParameter("cnt"));

    // <3> Compute Guess
    guess = (low+high) / 2;
    %>
    <form action="CheckResult.jsp">
        <h2>My Guess no. <%=++cnt %> is <%=guess %></h2>
        <input type="radio" name="guesser" value="too_low" onclick="submit()">
Too Low
        <input type="radio" name="guesser" value="too_high" onclick="submit()">
Too High
        <input type="radio" name="guesser" value="perfect" onclick="submit()">
Perfect

        <input type="hidden" name="low" value="<%=low %>"/>
        <input type="hidden" name="high" value="<%=high %>"/>
        <input type="hidden" name="cnt" value="<%=cnt %>"/>
    </form>
</body>
</html>
```

CheckResult.jsp

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1"%>
<!DOCTYPE html>
<html>
<body>
    <%
        //Fetch all variables
        int low = Integer.parseInt(request.getParameter("low"));
        int high = Integer.parseInt(request.getParameter("high"));
        int cnt = Integer.parseInt(request.getParameter("cnt"));
        int guess = (low+high) / 2;

        //Get the value of radio button
        String check = request.getParameter("guesser");
```

```
        if(check.equals("perfect"))
        {
%>
            <jsp:forward page="Winner.jsp">
                <jsp:param value="<%=cnt %>" name="cnt"/>
            </jsp:forward>
        <%
        }
        else if(low>=high)
        {
%>
            <jsp:forward page="Cheater.jsp"/>
        <%
        }
        else
        {
            if(check.equals("too_low"))
            {
                low = guess;
            }
            else
            {
                high = guess;
            }
        }
%>
        <jsp:forward page="NextGuess.jsp">
            <jsp:param value="<%=low %>" name="low"/>
            <jsp:param value="<%=high %>" name="high"/>
            <jsp:param value="<%=cnt %>" name="cnt"/>
        </jsp:forward>
    <%
    }
%>
</body>
</html>
```

Winner.jsp

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1"%>
<!DOCTYPE html>
<html>
```

```
<body>
    <h1>I Guessed in <%=request.getParameter("cnt") %> Attempts</h1>
    <a href="Index.jsp">Play Again</a>
</body>
</html>
```

Cheater.jsp

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1"%>
<!DOCTYPE html>
<html>
<body>
<h1>You are a Cheater</h1>
</body>
</html>
```

Output:

Number Guesser

Select a Number between 0 and 100

Start

lets assume 93

My Guess no. 1 is 50

☐ Too Low ☐ Too High ☐ Perfect

My Guess no. 2 is 75

☐ Too Low ☐ Too High ☐ Perfect

My Guess no. 3 is 87

☐ Too Low ☐ Too High ☐ Perfect

My Guess no. 4 is 93

☐ Too Low ☐ Too High ☐ Perfect

I Guessed in 4 Attempts

[Play Again](#)

For Cheating

You are a Cheater